

THE UMass X-Pol MOBILE DOPPLER RADAR: DESCRIPTION, RECENT OBSERVATIONS, AND NEW SYSTEM DEVELOPMENTS

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ABSTRACT

Accurate detection and forecasting of severe storms has driven radar tornado probing as an important research topic that has added significantly to the understanding and prediction of severe weather phenomena. The need for close range observation, coupled with the fact that significant information is buried at low altitudes, has made mobile Doppler radars an important tool in the characterization of severe storms and related weather phenomena. The UMass XPol radar is one such truck mounted dual-polarized Doppler radar operating at 9.41 GHz.

This paper documents the existing mobile radar system and presents close range, high resolution observations of Doppler velocity, reflectivity, differential reflectivity and cross-polarization correlation coefficient of a variety of storms observed during 2007. Additionally, the architecture of a planned pulse compression system upgrade is described.

Index Terms— Dual-polarized, reflectivity, differential reflectivity, cross-polarization correlation co-efficient, pulse compression.

1. INTRODUCTION

The need for accurate observation of severe storms, along with scientific research characterizing their nature and subsequent incorporation into numerical models, has driven the development of mobile Doppler radars. This, coupled with the small spatial scales of tornadic events and their proximity to ground level, necessitates close range observation. Dual-polarized radars have enhanced our understanding of tornadic supercells by providing valuable information about the shape of the scatterers and more recently, classification of hydrometeors into rain, hail or graupel [1-2]. The University of Massachusetts Microwave Remote Sensing Laboratory (MIRSL) collaborates with the

University of Oklahoma School of Meteorology (SoM) in the study of tornadoes and severe storms using mobile meteorological radars such as XPol [3].

The purpose of this paper is twofold. First, it presents close range observations of reflectivity, Doppler velocity, differential reflectivity and cross-polarization coefficient of two severe storms: an EF-5 tornadic supercell over Greensburg, KS on 4 May 2007 and very close range observations of another supercell on 13 May 2008. Second, a planned pulse compression system upgrade to XPol is documented. In addition to providing the usual benefits of increased range resolution and higher average transmitted power for tornado observation, the planned system upgrade is such that XPol will serve as a testbed for future radars under development by the CASA Engineering Research Center.

2. UMASS XPOL RADAR DESCRIPTION

The UMass XPol radar is a truck-mounted dual-polarized pulsed Doppler radar. Fig.1 shows an XPol deployment near Okeema, OK. The antenna is a 1.8 m parabolic dish with dual polarized feeds mounted on an elevation-over-azimuth pedestal. The transmitter is a Raytheon ST/MK2 high sea navigation radar. The microwave power source is a magnetron with a 0.1 % duty cycle and is housed in a white box in the truck trailer, as seen in Fig.1. The conventional coherent-on-receive technique is employed for estimation of phase-derived products. The entire design is built around a Linux based host computer that is housed in the truck cab. The construction of the radar is such that it is fully operable from inside the truck.

Significant radar parameters of XPol are listed in Table 1. The operator can select between 4 different pulsing schemes that the computer communicates to an FPGA controller. In the presented datasets, the first staggered pulse repetition frequency is used to increase the maximum unambiguous



Fig. 1: The mobile dual-polarized radar XPol being deployed near Okeema, OK on May, 13th 2008.

velocity range whilst also maintaining an adequate unambiguous range. Furthermore, the operator can choose between automating the scan by entering the scan parameters on the computer or do the same manually by using the positioner controller inside the truck.

TABLE I: XPOL RADAR CHARECTERISTICS [3]

ANTENNA	
Gain	41 dB
3-dB Beamwidth	1.25°
Type	Dual-polarized parabolic reflector
TRANSMITTER	
Peak Power	6 kW for H and V each
Frequency	9.41 GHz
PRF	1.6 kHz, 2.4 kHz, 510.2 Hz stagger 4.8 kHz, 3.2 kHz, 403.8 Hz stagger 1 kHz 500 Hz
RECEIVER	
Dynamic range	70 dB
Min. detectable signal	-5 dBZ at 10 Km
IF	60 MHz
Noise figure	4 dB

3. RECENT OBSERVATIONS

Polarimetric observations of an EF-5 tornado over Greensburg, KS on May 4th, 2007 are presented in Fig. 2. The fateful tornado that wiped out the entire city of Greensburg, KS was reported at 8:55 PM local time [4], 37 minutes after the dataset presented below was recorded. These images were recorded from a location near Protection, KS approximately 25 Km from the mesocyclone. Features at near ranges are second trip echoes, as they exhibit random velocities. This is a result of the random pulse-to-pulse phase of the magnetron source.

Fig. 3 presents very close range observations of a supercell near Okeema, OK. In this case the radar was optimally

placed to observe the developing tornado from the “clear slot”. Although a well defined hook echo was observed, no tornado touched down in this case.

4. SYSTEM UPGRADES

As mentioned earlier, the primary motivation for pulse compression is increased range resolution for a given transmit duration. XPol currently operates with a fixed range resolution of 150 m, and is going through a pulse compression upgrade. To this end, a traveling wave tube amplifier (TWTA), with a peak power rating of 250 W, is to replace the magnetron. The lower peak power rating of the TWTA is intentional, to align the XPol implementation to those of solid-state CASA network radars under separate development. This serves to evaluate sensitivity issues associated with the implemented pulse compression scheme.

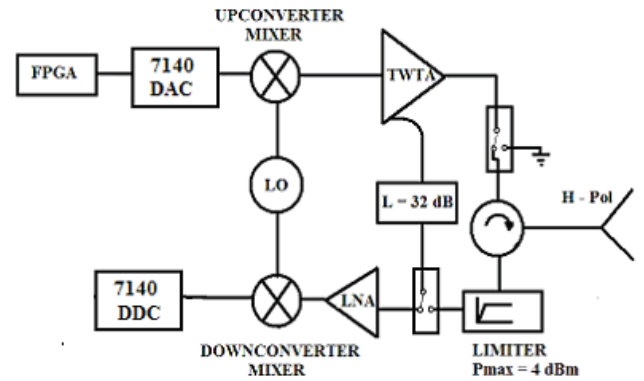


Fig. 4: Proposed architecture for implementation of Pulse Compression in XPol

Fig. 4 shows a schematic diagram of the proposed Pulse compression architecture, where only one of the two polarizations (H) is shown for ease of illustration. The proposed architecture involves the use of a Direct Digital System, (DDS), to generate a frequency sweep. This then is upconverted to X-Band using a single sideband upconverter, and fed as the input to 250 W peak power Traveling Wave Tube Amplifier (TWTA). Because the TWTA is a CW tube, care must be taken to suppress the transmitted noise between pulses. A switching arrangement is thus employed to circumvent the noise from the high gain TWTA. The FPGA serves to both switch the DDS ON and OFF, and also to control the switches at the end of the transmit and receive cycles. The switches, in conjunction with a directional coupler and attenuator, also aid in recording the transmitted signal through a separate calibration loop. The Local Oscillator is tuned to a frequency of 9.47 GHz and the gain of the LNA is 40 dB. After downconversion, the IF is fed to a digital receiver, Pentek 7140. The resultant baseband I and Q signals will be digitized and stored raw in a RAID that is housed in the Linux based host computer.

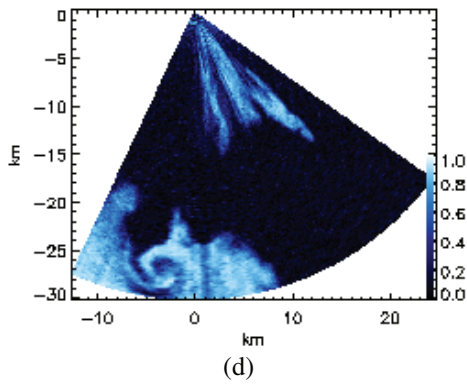
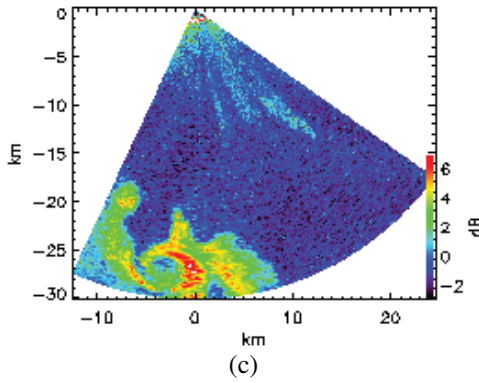
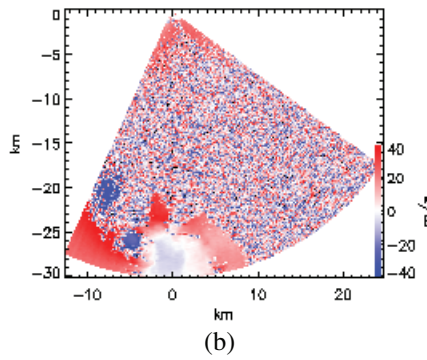
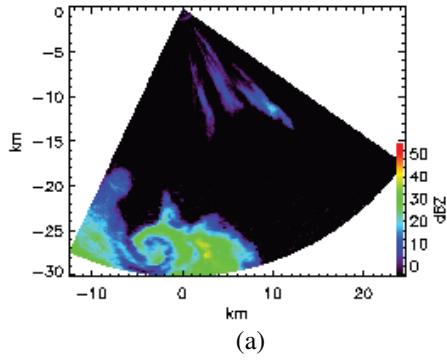


Fig. 2: Doppler radar images of an EF-5 tornado obtained near Greensburg, KS at 8:18 PM local time on May 5th, 2007. XPol was operated at 7° elevation. (a) Reflectivity (b) Doppler Velocity (c) Differential Reflectivity (d) Cross-Polarization Correlation

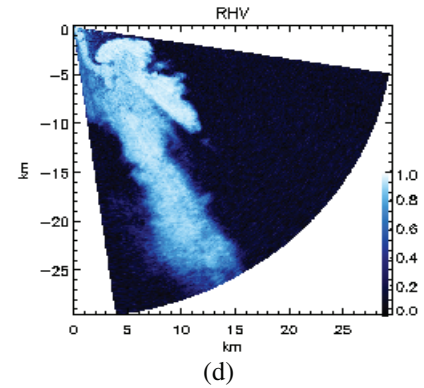
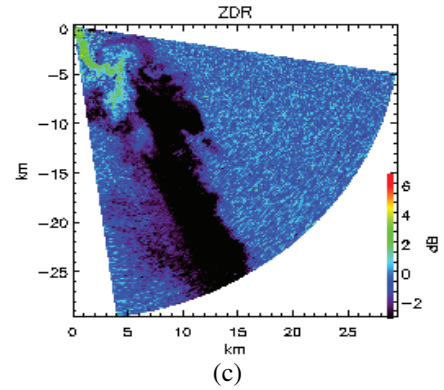
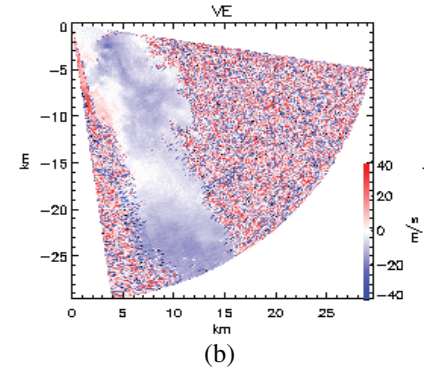
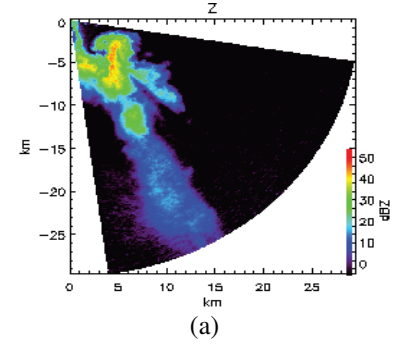


Fig. 3: Doppler Radar images of a supercell near Okeema, OK at 7:36 PM local time on May 13th, 2008. Xpol was operated at 6.5° elevation. (a) Reflectivity (b) Doppler Velocity (c) Differential Reflectivity (d) Cross Polarization Correlation

An IDL routine will then process the raw data, and display significant meteorological parameters of interest in near real time.

5. FUTURE WORK

Targets distributed in range, such as extended precipitation systems, tend to corrupt weak events occurring near regions of strong echoes. This necessitates particularly low range-time sidelobe levels. Although it has been established that sidelobe suppression filters can bring them down to acceptable levels for research purposes [5], it is also true that precious transmitted power is discarded in the process. In this context, challenges in the form of optimal choice of weighting functions for mismatch filter design still lie ahead for XPol. The performance of linear frequency modulated and various non-linear frequency modulated waveform designs will be evaluated by interpreting the Doppler velocity and Reflectivity estimates from XPol during precipitation in light of the ambiguity function. Additional information can be obtained by comparing velocity estimates from the stationary ground clutter, to the ambiguity function evaluated at zero Doppler.

6. REFERENCES

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